

Alex Christopher

AI Engineer — Computer Vision & Data Automation

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SUMMARY

AI engineer with production experience building computer vision and LLM-agent systems that ship into real operations, not just notebooks. Over the past year and a half I've taken projects from a manual, error-prone process to an automated system a non-technical team relies on daily — cutting a multi-minute manual check down to seconds, catching a compliance risk before it reaches print or a customer, and giving business teams a way to ask their own data a question in plain language instead of waiting on a report. I measure my work by the business outcome it produces, not just the model behind it — every project below is described the same way: what the problem cost before, what I built, and what changed after. Open to relocating to Malaysia.

EXPERIENCE

AI Engineer — PT. Wings Surya

Sep 2025 – Present

Surabaya, Indonesia

- Design and ship computer vision and LLM-agent systems end to end — from model and pipeline design through to a tool that non-technical retail, supply-chain, and warehouse teams use every day. Five of these systems are detailed under Selected Projects below, with the problem they solved and the measured impact.
- Work directly with business stakeholders (sales, SCM, trade-marketing) to translate a manual, spreadsheet- or eyeball-driven process into an automated one — starting from how the team actually works today, not just the technical spec.
- Keep deployed systems cheap to run and adopt: a single CPU server, the warehouse's existing CCTV feed, and the team's existing Google Sheets workflow, rather than new hardware, GPU clusters, or a tool nobody asked for.

AI Engineer — PT. Visiniaga Mitra Kreasindo

Jan 2025 – Aug 2025

AI R&D — scoped, built, and delivered AI project work for external clients across different requirements and problem domains.

SELECTED PROJECTS

Agentic AI for ERP — Telegram Bot, Text-to-SQL **FEATURED**

One of the businesses I support runs its operations on an ERP database that only the finance/data team could query directly — everyone else waited on a report. I built a Telegram-based LLM agent that lets anyone ask a business question in plain language: it writes a safe, read-only SQL query, reasons over the result against real business caveats (e.g. avoiding double-counting a rolling sales ledger), and replies with numbers and an auto-generated chart in the same chat — extended to work across AR, AP, Sales, Purchasing, Logistics, and Inventory, not a single-purpose bot.

Impact: turned a multi-day report request into a few minutes of back-and-forth chat. In one real session, a follow-up question surfaced a set of products that looked like top sellers by revenue but were quietly running negative margin — the kind of finding that used to only surface at quarter-end close, if at all.

Stack: LLM Agent, Text-to-SQL, Python, SQL, Chart Rendering, Telegram Bot API

Computer Vision KPI Scoring — YOLO Planogram Compliance **FEATURED**

Sales reps checked freezer planogram compliance by eye, store by store — a 5-7 minute manual walk-through per visit, with no consistent record of what was actually found. I trained a YOLOv11 model that reads a single top-down photo and scores three things — how full the freezer is, how tidy it's arranged, and how much of the SKU catalog is present — derived directly from two detection passes (compartment status, then SKU identification), with no manual review step in between.

Impact: cuts the check to about 15 seconds — roughly 96% faster — running on a single CPU server, no GPU cluster or per-image API cost. At the scale of a national freezer network, that's illustratively worth several reps' worth of time freed every month: redirectable to selling time, or to covering more outlets without adding headcount.

Stack: YOLOv11, Computer Vision, Python, OpenCV, KPI Scoring

Real-Time Pallet Fullness Detection — Warehouse CCTV

Every pallet entering a warehouse is assumed to be full — if an incomplete one gets logged as stock unnoticed, the next stock count comes up short, and if that pallet was tied to a customer order, the shortage becomes a complaint. I deployed a YOLO model on the warehouse's existing CCTV feed, no new camera hardware, that classifies every pallet as full or not-full in real time and tracks it against a virtual boundary line marking the warehouse entrance.

Impact: when an incomplete pallet crosses that line, the system fires an alarm before it's ever counted as stock — catching the exact scenario that turns into a customer complaint (say, 100 boxes ordered, 98 shipped) at the cheapest possible point to fix it, without needing a supervisor stationed at the door every shift.

Stack: YOLO, Computer Vision, Real-time Detection, RTSP/CCTV, Python

Packaging Design Diff Tool — Computer Vision QA

Every packaging revision at a company running thousands of SKUs a year needs a human to hold the old proof next to the new one and catch every change — a price tag added, a net weight updated, a ribbon recolored. Miss one and it goes to print. I built an OpenCV pipeline that aligns two design files (correcting for scan/export drift), runs a structural diff tolerant of compression noise, and draws a bounding box around every region that meaningfully changed.

Impact: turns "compare these two images" into "look at these three highlighted spots" — built to catch the kind of change a reviewer's accuracy naturally misses after the twentieth near-identical proof in a row, before a reprint or a wrong number on shelf happens.

Stack: Computer Vision, OpenCV, Image Diffing, Python

New Product Demand Forecasting Automation — BigQuery/SAP → Sheets + AI Bot

With dozens of new products launching every month, the Supply Chain team calculated each one's demand forecast by hand in Excel — pulling data, applying the formula, updating the sheet, every SKU, every month. I automated the pipeline: a script pulls source data from BigQuery (sourced from SAP) and fills a Google Sheet using SCM's own formula, tracking each product through its first two months on shelf against that estimate. On top of that, I connected an AI bot so anyone can ask a natural-language question — e.g. "the 5 worst-performing completed products" — and get a filtered, ranked answer instead of scrolling a 40-plus-row sheet.

Impact: removed the manual monthly Excel update entirely and made the calculation consistent — the same formula runs the same way every time, no transcription drift. The bot layer also caught a pattern a quick glance would miss: 28 "completed" products still showing month-2 demand above 100% of estimate, meaning finished status didn't mean repeat demand had actually tapered off.

Stack: Python, BigQuery, Google Sheets API, LLM Agent, SAP

SKILLS

AI / LLM

LLM Agents, Text-to-SQL, Prompt Engineering, RAG, OpenClaw, Hermes Agent

Computer Vision	YOLOv11, OpenCV, Object Detection, Image Diffing, CVAT
Data Engineering	BigQuery, SQL, ETL Automation, Google Sheets/Excel API
Fullstack	Python, TypeScript, React, REST APIs, C#
Tools & Infra	Docker, Git, Vercel, Linux, Claude Code, ChatGPT Codex

EDUCATION

B.Eng, Informatics Engineering — Universitas Kristen Petra

2015 - 2019